

ECE 313: Probability with Engineering Applications

2025 Fall Instructors: Piao Chen & Xu Chen

Homework 10

Name: _____

Student ID: _____

Due Date: December 19 23:59, 2025

Problem 1. Suppose X and Y are jointly Gaussian with $\mu_X = 2$, $\mu_Y = 1$, $\sigma_X^2 = 4$, $\sigma_Y^2 = 1$, and $\rho_{XY} = -\frac{1}{4}$. Let $W = X + aY + b$. Find the values of a and b that make X and W uncorrelated. Will these values of a and b make X and W independent? Explain why.

Problem 2. Suppose Y and W are jointly Gaussian random variables with $E[Y] = 2$, $E[W] = 0$, $\text{Var}(Y) = 16$, $\text{Var}(W) = 4$, and $\rho = 0.25$. Let $X = 3Y + W + 3$.

- (a) Find $E[X]$ and $\text{Var}(X)$.
- (b) Calculate the numerical value of $P\{X \geq 20\}$.
- (c) Find $E[Y|X]$. Your answer should be a function of X .

Problem 3. Suppose $X \sim \text{Unif}(1, 2)$, and given $X = x$, Y is exponential with parameter $\lambda = x$.

- (a) Find $\text{Cov}(X, Y)$.
- (b) Let $Z = X^2$. Find $\text{Cov}(X, Z)$.

Problem 4. The following parts are independent.

- (a) Consider random variables X and Y on the same probability space. If $\text{Var}(X + 2Y) = 40$ and $\text{Var}(X - 2Y) = 20$, what is $\text{Cov}(X, Y)$? If you are also told that $\text{Var}(X) = \text{Var}(Y)$, what is $\rho_{X,Y}$?
- (b) If $\text{Var}(3X + Y) = \text{Var}(3X - Y)$, are X and Y uncorrelated?
- (c) If $\text{Var}(X) = \text{Var}(Y)$, are X and Y uncorrelated?